



IoT-Based Home Automation

Guide

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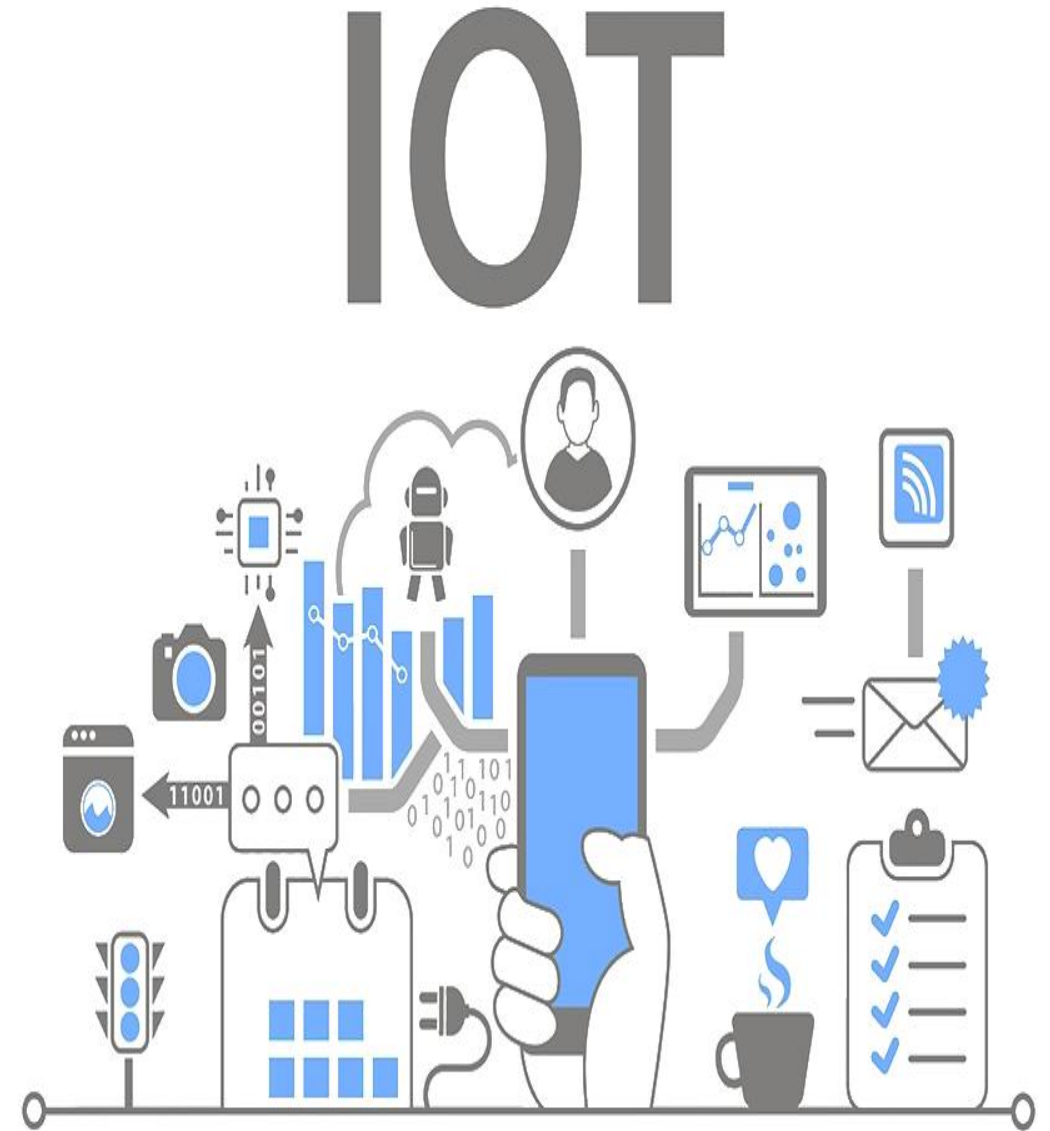
10. References and Final Conclusion

References we found from the internet and our final conclusion or learnings we learn from this project



Introduction

- ❑ An network of physical objects—“things”—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.



Overview of Project

- ❑ We found many IOT Based projects such as Ambient Light on Arduino, Home Curtain Systems, Smart Kitchen, Home security systems, connections over household electronic appliances, etc. But finally, we decided to make an Automatic Home Curtain System or Smart Curtain System with some new features.
- ❑ Smart Curtain System is a system that works based on humans' commands. We have to give the commands using the mobile to this system, and according to our commands, it will give the response.

Literature Reviews

Understood IoT, Hardware, and software we need

Where we knew what is IoT and how it works! Also, know about some important hardware and software

Learned about Blynk

Where we learned to use Blynk and also learned connect our ESP8266 with the Blynk Cloud Platform

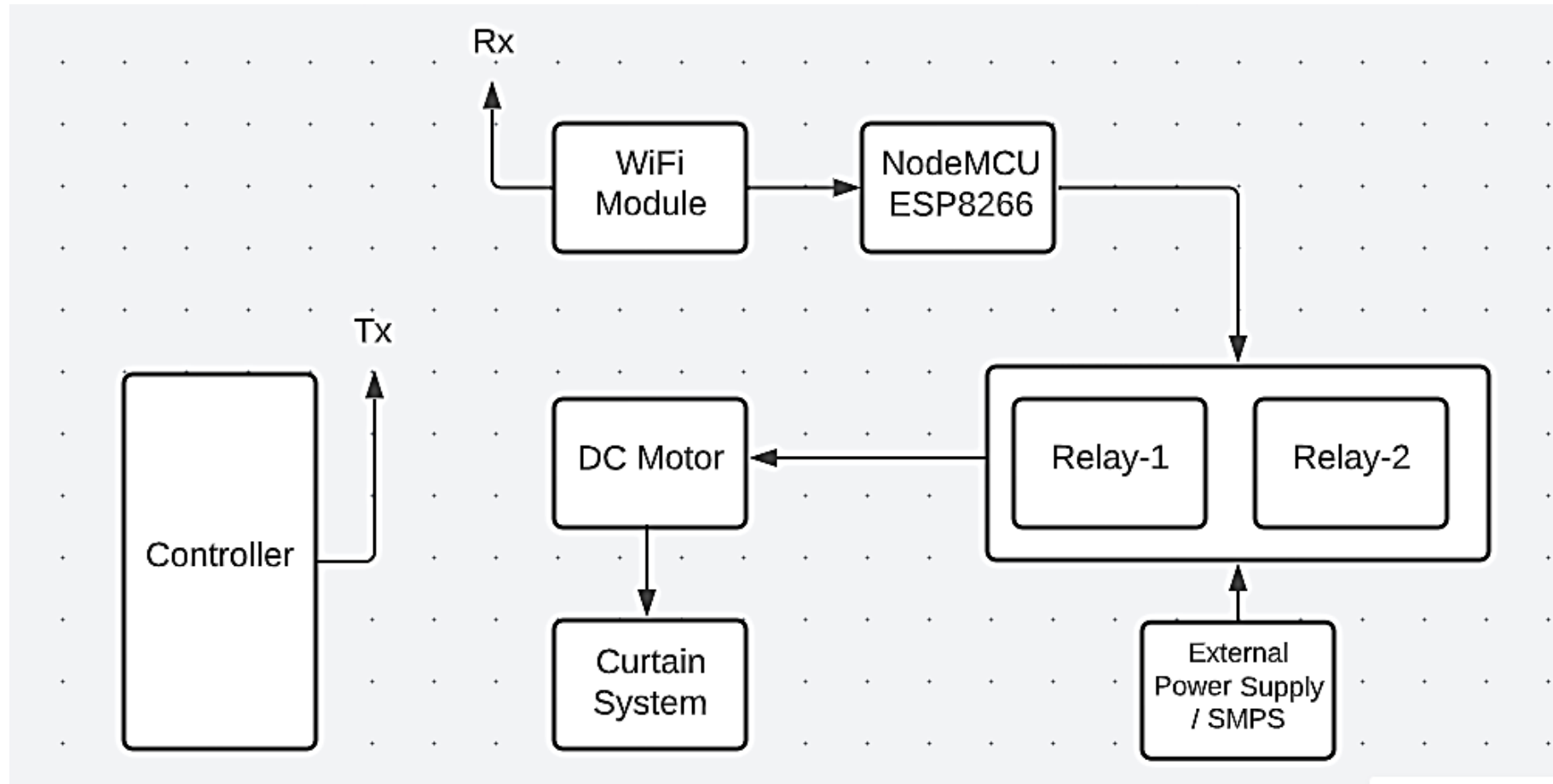
Setup and Programming

Where we learned to setup and program our ESP8266 in Arduino IDE

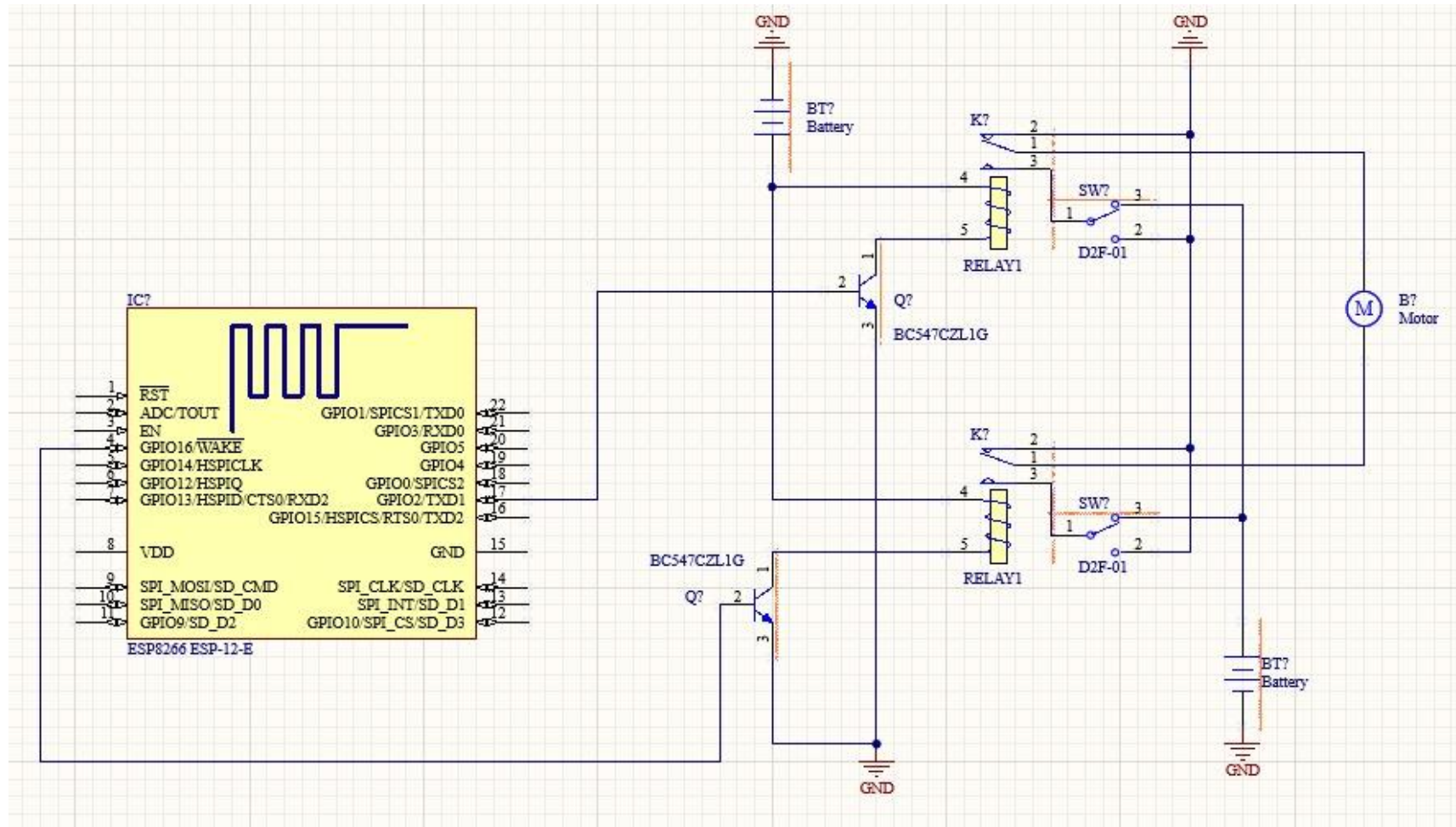
Tinker cad for simulation

To know how our circuit works and behaves, we use the way that is an online simulation, And for that we used Tinker cad!

The Block Diagram Of Our IoT System



Basic Circuit Diagram of Our IoT System



Required Electronics Components

(Hardware)

NodeMCU ESP8266

- ❑ ESP8266 is a low-cost wifi microcontroller with a full TCP/IP stack. It has many GPIO (general purpose input output pins) for interfacing with different sensors.
- ❑ ESP8266 due to its low cost and good functionality is used a lot in prototyping IoT products.
- ❑ Its cross-functionality with Arduino makes it easy to program with Arduino IDE.



2-Channel Relay Module

- ❑ A relay is a simple electromechanical switch. While we use normal switches to manually close or open a circuit, a Relay is also a switch that connects or disconnects two circuits.
- ❑ But instead of a manual operation, a relay uses an electrical signal to control an electromagnet, which in turn connects or disconnects another circuit.



+24V DC Motor

- ❑ A DC motor is defined as a class of electrical motors that convert direct current electrical energy into mechanical energy.
- ❑ When electric fields and magnetic fields interact, a mechanical force arises. This is the principle on which the DC motors work.



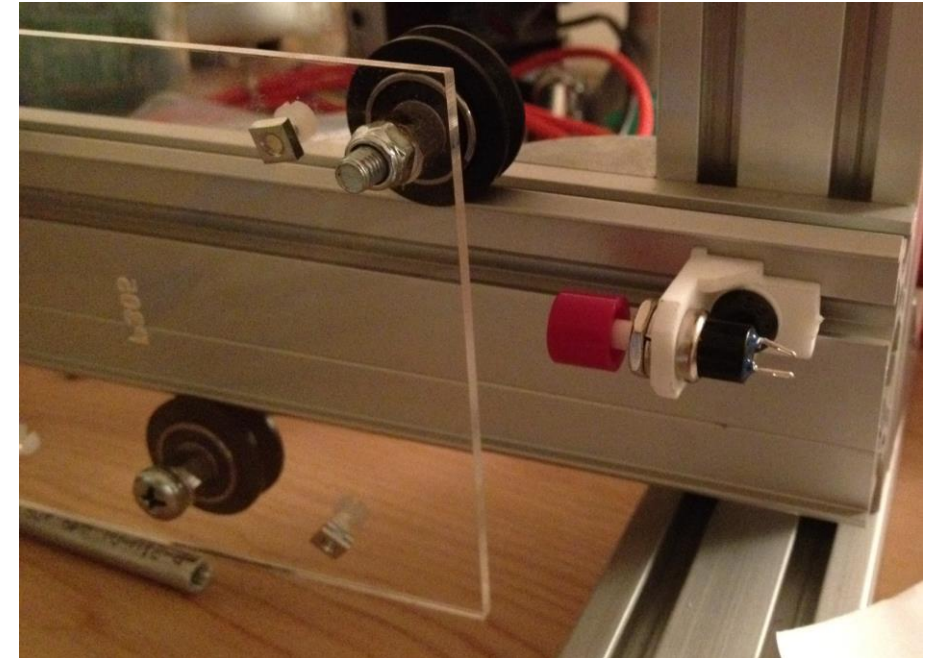
Switched Mode Power Supply

- ❑ A switching power supply (SMPS) is an electronic circuit that converts power using a switching device that is turned on and off at a high frequency.
- ❑ A storage device such as an inductor or capacitor to provide power when the switching device is not available.



Limit Switch (Plunger)

- ❑ A limit switch is an electronic device in which a physical force is applied to an object.
- ❑ A limit switch is used to determine the presence or absence of an object.
- ❑ These switches are used to set the limit of movement of objects, so they are called Limit Switches.



Required Software & Applications

(Software)

Required Software & Applications



Arduino IDE

For Programming / To
program the NodeMCU
ESP8266



Blynk

Web-Based or Android-
Based Software



Multisim Live

Web-Based Circuit
Simulator

Arduino IDE

- ❑ The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. This software can be used with any Arduino board.
- ❑ Using Arduino IDE, we program our NodeMCU ESP8266.



Blynk

- ❑ Blynk is a Platform with iOS and Android apps to control Arduino, Raspberry Pi, and the likes over the Internet.
- ❑ It's a digital dashboard where you can build a graphic interface for your project by simply dragging and dropping widgets.
- ❑ You can use it Web-Based and also as an Android-Based Application.



Multisim Live

- ❑ Multisim™ Electronics Workbench™ is industry-standard SPICE simulation and circuit design software for analog, digital, and power electronics in education and research.



Benefits

Convenience

Better Home
Management

Complete Control



Drawbacks



Security Issues



Cost: Extremely expensive



Maintenance



Real World Applications



Lighting control



Household electronic
Appliances



AI-driven digital
experiences

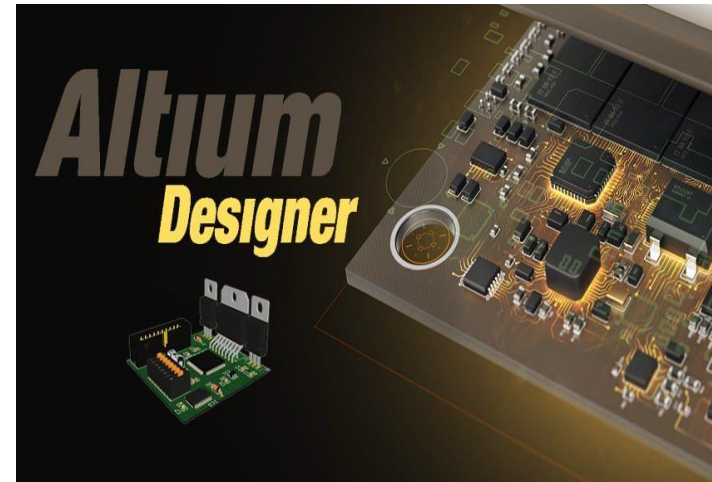


Online Simulation

Circuit Simulation



PCB Layout of our project



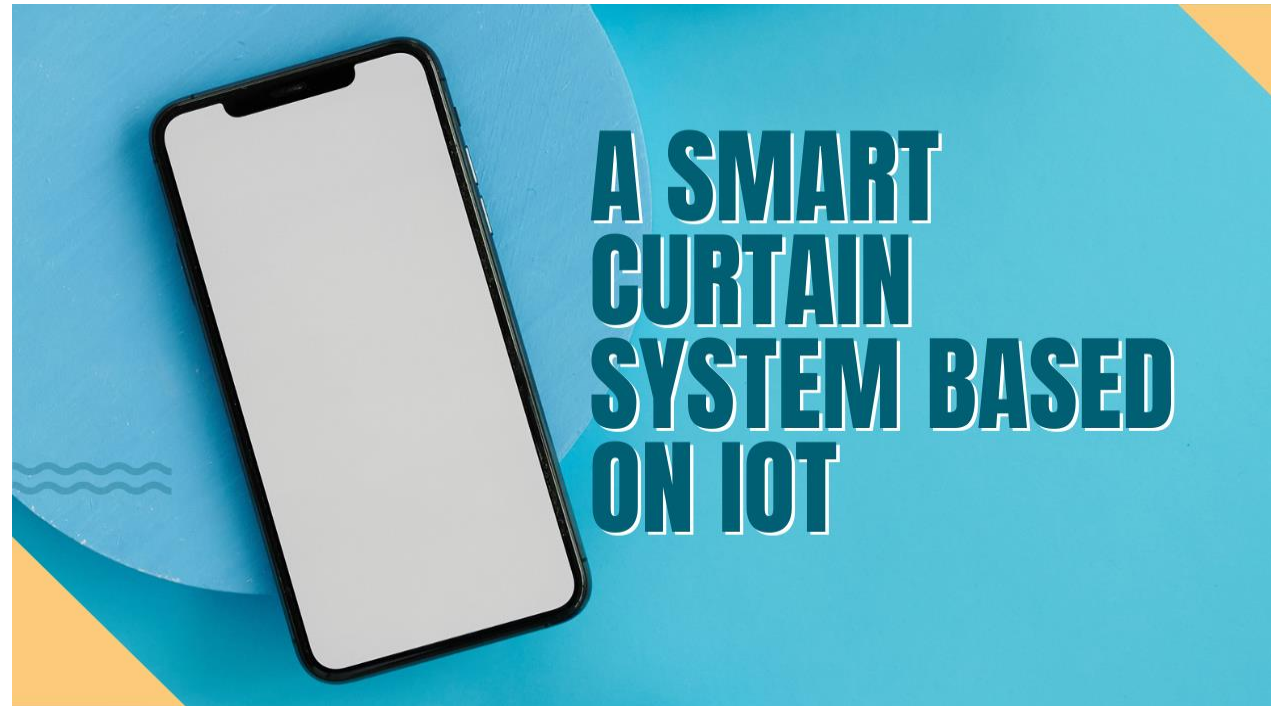


Let's See The PCB Layout

Our Future Goals

- ❑ To make a proper IoT-based curtain system connected to the internet. By connecting this to the internet we can access this system from any part of the world via the internet. To make such a system, we have to connect this system with the router, and the router connects it to the internet.

Video of Working Model



Conclusion

- ❑ By doing this project we learn about various aspects of electronics. We learn how to make a circuit. This made us learn the analog aspect of our project. We program the microcontroller. This was the embedded aspect. We used ESP 8266 Wi-Fi chip to control the motor. That was the network side of our project & we assembled this project with a mechanical structure, this thing made us learn the mechanical side also.
- ❑ We also came up with some new ideas and we are going to apply that ideas to our project in the next term of our project.

Acknowledgment

- ❑ For now, we only wish to simply acknowledge Mr. Ketan S. Patel, who helps us during this entire project.

References

[To know more about NodeMCU ESP8266](#)

[To learn to program NodeMCU on Arduino IDE](#)

[To learn more about Relay](#)

[To learn to control ESP8266 with Blynk Cloud Application](#)

[Setting up the Blynk with ESP8266](#)



References

[Download Blynk IoT Android Application](#)

[Guide to setup and program the NodeMCU ESP8266](#)

[Learn to connect ESP8266 with Blynk \(Web-Based and App-Based \)](#)

[Complete Guide to do IoT-Based Home Automation with Blynk](#)

[IoT Home Automation using ESP8266 with Blynk](#)



Summary

✓ Introduction and Literature Reviews

We saw the basic introduction about home automation, IoT and about our project

✓ Electronics Components

We discussed all the hardware parts and software that we will use to complete this project

✓ Block Diagram and Circuit Diagram

We saw the simple block diagram of our IoT-Based System and also saw the circuit diagram

✓ Advantages, Disadvantages and Real World Applications

Then we discussed the advantages, disadvantages and application of the project

✓ Simulation

And at the end, we successfully run both online and offline simulations

✓ Model

The video of the working model



Many thanks to you for giving your valuable time. We look forward to having you again in our next time. Your presence here definitely enriched this presentation.

And for this, we are grateful!